

MATH 2551-K MIDTERM 1
VERSION A
FALL 2023
COVERS SECTIONS 12.1-12.6, 13.1-13.4, 14.1

Full name: _____

GT ID:_____

Honor code statement: I will abide strictly by the Georgia Tech honor code at all times. I will not use a calculator. I will not reference any website, application, or other CAS-enabled service. I will not consult with my notes or anyone during this exam. I will not provide aid to anyone else during this exam.

() I attest to my integrity.

Read all instructions carefully before beginning.

- Print your name and GT ID neatly above.
- You have 75 minutes to take the exam.
- You may not use aids of any kind.
- Please show your work.
- Good luck! Write yourself a message of encouragement on the front page!

For problems 1-2 choose whether each statement is true or false. If the statement is *always* true, pick true. If the statement is *ever* false, pick false. Be sure to neatly fill in the bubble corresponding to your answer choice.

1. Any plane has only two distinct normal vectors.

☐ **TRUE**
☐ **FALSE**

2. If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^3$, then $\mathbf{u} \times \mathbf{v} + \mathbf{v} \times \mathbf{u} = \mathbf{0}$.

☐ **TRUE**
☐ **FALSE**

3. Order the following curves in increasing order of curvature.

- A) A helix of curvature 2
B) A line segment of length 10
C) A circle of radius 20

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4. Which of the following vectors could be the principal unit normal vector at time $t = 2$ to a curve whose tangent line at $t = 2$ is given by

$$\ell(p) = \langle 1, 0, -1 \rangle + p \langle 1, 2, 1 \rangle.$$

You must justify your answer to receive full credit.

- A) $\langle 1, 1, 1 \rangle$
B) $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$
C) $\langle -2, 1, 0 \rangle$
D) $\left\langle \frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0 \right\rangle$

5. Tobert, a hard-working documentarian, is hiking through the African savannah for two days and his path is given by the curve $\mathbf{r}(t) = \langle t^2, \frac{1}{3}t^3, 2 \rangle$ for $0 \leq t < 48$, where t is measured in hours and $\mathbf{r}(t)$ in meters.

- (a) What is Tobert's position two hours into his hike?
- (b) Find a function $s(t)$ that gives the total distance Tobert has hiked for any time t since he started at $t = 0$.
- (c) How far has Tobert hiked when he is at the point $(4, 8/3, 2)$?

6. In this problem, you will work with the differential equation

$$\mathbf{r}''(t) = 3\mathbf{i} + e^t\mathbf{j} + 49e^{7t}\mathbf{k}, \quad -\infty < t < \infty.$$

- (a) Find all vector-valued functions $\mathbf{r}(t)$ with $-\infty < t < \infty$ which are solutions to this equation.

Hint: Your answer to this part should include some undetermined constant vectors.

- (b) Find all vector-valued functions $\mathbf{r}(t)$ with $-\infty < t < \infty$ which are solutions to this equation and also have the property that $\mathbf{r}'(0) = 3\mathbf{j} + 7\mathbf{k}$.

- (c) Find all vector-valued functions $\mathbf{r}(t)$ with $-\infty < t < \infty$ which are solutions to this equation, have the property that $\mathbf{r}'(0) = 3\mathbf{j} + 7\mathbf{k}$ and also have the property that $\mathbf{r}(0) = \langle 1, 1, 5 \rangle$.

7. Let p_1 be the plane defined by the equation $2x - 3y + z = -10$ and let Q be the point $(3, 1, 1)$.

(a) Find an equation for the plane p_2 which contains the point Q and is parallel to p_1 .

(b) Find an equation for the line ℓ which passes through Q and is orthogonal to both planes.

(c) Find the point R where the line ℓ intersects the plane p_1 .

(d) Compute the distance between the two planes using your work in parts (a)-(c) above.

8. (a) Find and sketch the domain of the function $f(x,y) = \sqrt{4 - x^2 - y^2}$. Show your work and label your axes clearly.

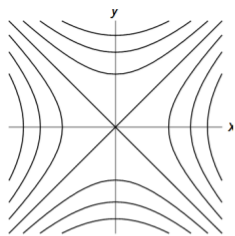
(b) Match the contour plots and graphs below with the given functions of two variables.

Function	Graph	Contour Plot
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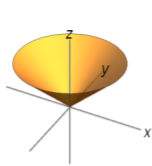
$f(x,y) = \cos\left(\sqrt{x^2 + y^2}\right)$	_____	_____
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$g(x,y) = x^2 - y^2$	_____	_____
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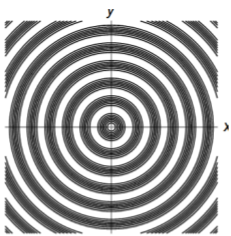
$h(x,y) = x^2 + 2y^2$	_____	_____
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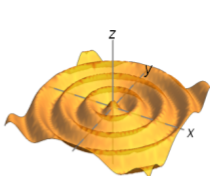
A



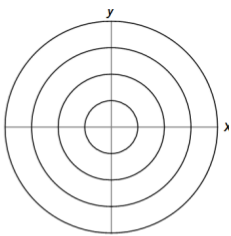
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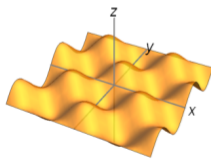
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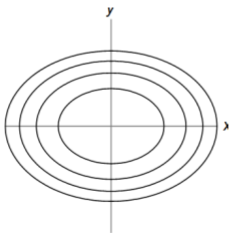
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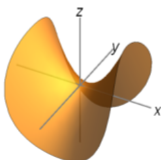
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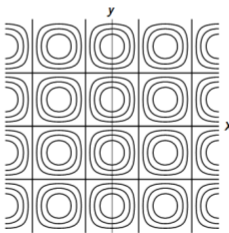
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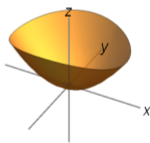
G



H



I



J

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FORMULA SHEET

- $\langle u_1, u_2, u_3 \rangle \cdot \langle v_1, v_2, v_3 \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3$

- $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos(\theta)$

- $\langle u_1, u_2, u_3 \rangle \times \langle v_1, v_2, v_3 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$

- $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| |\sin(\theta)|$

- $L = \int_a^b |\mathbf{r}'(t)| \, dt$

- $s(t) = \int_{t_0}^t |\mathbf{r}'(T)| \, dT$

- $\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{d\mathbf{r}}{ds}$

- $\kappa = \left| \frac{d\mathbf{T}}{ds} \right| = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right| = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$

- $\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$

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